

Futureproof

Equity Preservation Mortgage v14c Investment Opportunity Overview

Quantitative Risk Analysis | April 2025

April 2025

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Executive Summary

FutureProof has developed an innovative financial product — the Equity Preservation Mortgage (EPM) — that allows homeowners to access their home equity while retaining full ownership. The v14c quantitative model provides the actuarial basis for the product; an independent companion paper ("Model Assumptions and Parameter Risk", April 2025) quantifies the parameter uncertainty around the headline results.

Headline results are reported across three empirically-anchored scenarios, consistent with the companion paper:

Scenario	Basis	PoD (Yr 30)	Mean Surplus
Base case (model-as-written)	Current internal parameters	1.2%	\$2,435,992
Realistic-central	Empirically-centred μ , γ , collar	16.9%	~\$995,000
Adverse-plausible	Lower bounds on μ , γ , collar	43.2%	~\$253,000

The product remains insurable and commercially attractive across this range, but investors should understand that the base-case headline is one point on a distribution, not the distribution itself. Capital and reinsurance sizing are being calibrated to the adverse-plausible leg.

Key Investment Highlights

- **Large addressable market** — Homeowners aged 55-75 across AU, US, UK & NZ represent a combined \$15T+ equity pool. Current reverse mortgage penetration is <1% in all four markets.
- **Validated product economics at base case** — 50,000-path Monte Carlo confirms < 0.01% portfolio PoC at year 30 at base-case parameters; companion parameter-uncertainty paper anchors the realistic-central and adverse-plausible scenarios.
- **Unique run-off mechanism** — No voluntary prepayments or early terminations until all loan costs are paid from the investment account. This eliminates early exit risk that plagues traditional mortgage products.
- **Multiple revenue streams** — origination fees (1.5%), ongoing margin (0.25%), surplus sharing (50% of surplus at maturity), and insurance premium income.
- **Institutional partnerships** — BlackRock passive index-linked ETFs (S&P500) with Volatility Control, plus SpiderRock continuous dynamic hedging.
- **Defensible IP** — proprietary Payments Waterfall, holiday mechanism, run-off mechanism, equity preservation structure, and quantitative risk engine create significant barriers to entry.
- **Transparent risk disclosure** — all scenarios, confidence intervals, and parameter sensitivities are disclosed upfront to investors and reinsurers.

The Market Opportunity

The Problem — A Global Challenge

- Homeowners over 60 across developed economies hold trillions in residential property equity
- Most have limited income in retirement but are asset-rich — the "asset-rich, cash-poor" dilemma
- Traditional reverse mortgages have a negative reputation globally (compound interest, no upside, estate erosion)
- Home downsizing has significant emotional, social, and tax implications in every market
- No existing product preserves equity while generating retirement income

The FutureProof Solution

- Homeowner borrows against equity at up to 80% LVR — a calculated portion of loan proceeds held in mortgage offset account and invested by BlackRock in index-linked ETFs (predominantly S&P500)
- Investment returns pay the mortgage interest + generate an annuity income (\$25K/yr for 10 years)
- P&I; structure amortises the loan to zero over 30 years — creating a structural safety net
- Holiday mechanism protects during market downturns (pauses interest payments)
- Unique run-off mechanism — no early exit until investment account has covered all loan costs
- At maturity: surplus split 50/50 between FutureProof and Mortgage Funder
- Partial profit realisation every 5 years (10% of surplus drawn into portfolio)

Multi-Region Market Opportunity

FutureProof is designed from the ground up as a multi-region platform. The EPM product is applicable wherever homeowners hold significant property equity and face the retirement income challenge. Our technology platform already supports region-specific configuration for currency, LTV limits, regulatory requirements, and tax treatment across all four target markets.

United States — Primary Growth Market

The US represents the largest addressable market globally. Over 40 million homeowners aged 55+ hold an estimated \$12 trillion in home equity. The existing HECM (Home Equity Conversion Mortgage) program is government-backed but capped at \$1,149,825 — leaving the jumbo market entirely unserved. High-value homeowners in coastal markets (California, New York, Florida) are the ideal EPM demographic.

Metric	United States
Currency	USD (\$)
Target demographic	Homeowners 55+, home value \$500K-\$10M
Addressable households	5,000,000+
Residential equity pool	~\$12 trillion
Regulatory body	Consumer Financial Protection Bureau (CFPB)
Licensing	NMLS (Nationwide Multistate Licensing System)
Competitive landscape	HECM dominates; no private equity preservation product exists
Tax advantage	EPM income may be tax-advantaged under IRS guidelines
Max LTV	80%

Australia — Founding Market

Australia is FutureProof's founding market and the base for the v14c model validation. Australian homeowners over 60 hold approximately \$1 trillion in residential property equity. The reverse mortgage market is small (~\$3B outstanding) and declining due to negative perception. Major banks exited the space in 2018, creating a significant gap that EPM is uniquely positioned to fill.

Metric	Australia
Currency	AUD (A\$)
Target demographic	Homeowners 55-75, property value A\$500K-A\$10M
Addressable households	200,000+
Residential equity pool	~A\$1 trillion
Regulatory body	Australian Securities & Investments Commission (ASIC)
Licensing	Australian Financial Services Licence (AFSL)
Competitive landscape	Heartland Finance only major player; banks exited 2018
Tax advantage	EPM income generally tax-free as return of capital (ATO)

Metric	Australia
Max LTV	80%

United Kingdom — Established Equity Release Market

The UK has the most mature equity release market outside the US, with ~\$50B outstanding. However, existing products are traditional lifetime mortgages with compound interest. The UK market is well-regulated by the FCA and the Equity Release Council provides consumer standards. British homeowners hold approximately £3 trillion in property equity, with property values in London and the South East creating an ideal high-value demographic for EPM.

Metric	United Kingdom
Currency	GBP (£)
Target demographic	Homeowners 55+, property value £300K-£10M
Addressable households	1,500,000+
Residential equity pool	~£3 trillion
Regulatory body	Financial Conduct Authority (FCA)
Licensing	FCA Authorisation
Competitive landscape	Mature equity release market (~£50B); no equity preservation product
Tax advantage	May qualify for tax relief under HMRC guidelines
Max LTV	80%

New Zealand — Adjacent Market

New Zealand is a natural adjacency to the Australian market — similar legal framework, property market dynamics, and demographic profile. The reverse mortgage market is small but growing, with Heartland Group being the dominant player. NZ property values have risen significantly, creating the same asset-rich, cash-poor dynamic. FutureProof can leverage AU infrastructure and regulatory expertise for efficient NZ market entry.

Metric	New Zealand
Currency	NZD (NZ\$)
Target demographic	Homeowners 55+, property value NZ\$500K-NZ\$10M
Addressable households	80,000+
Residential equity pool	~NZ\$200B
Regulatory body	Financial Markets Authority (FMA)
Licensing	Financial Advice Provider Licence (FAP)
Competitive landscape	Heartland dominates; minimal competition
Tax advantage	Treatment under IRD guidelines — consult tax advisor
Max LTV	80%

Combined Market Sizing

Market	Addressable Households	Equity Pool	At < 0.01% Penetration
United States	5,000,000+	\$12T	\$60B portfolio
Australia	200,000+	A\$1T	A\$5B portfolio
United Kingdom	1,500,000+	£3T	£15B portfolio
New Zealand	80,000+	NZ\$200B	NZ\$1B portfolio
Combined	6,780,000+	\$15T+ (USD equiv.)	\$75B+ portfolio

Even at conservative 0.5% penetration across all four markets, the combined portfolio opportunity exceeds \$75 billion in AUM — generating approximately \$1.5 billion in annual revenue across all income streams. The US market alone represents 80% of the addressable opportunity.

Go-To-Market Strategy

- **Phase 1 (Year 1-2):** Launch in Australia — pilot 50-100 EPMS, validate operations, refine pricing
- **Phase 2 (Year 2-3):** Scale Australia to 500 EPMS; enter New Zealand leveraging AU infrastructure
- **Phase 3 (Year 3-5):** Enter UK market — largest equity release market with clear regulatory pathway (FCA)
- **Phase 4 (Year 4-6):** Enter US market — largest opportunity; establish NMLS licensing, target jumbo segment

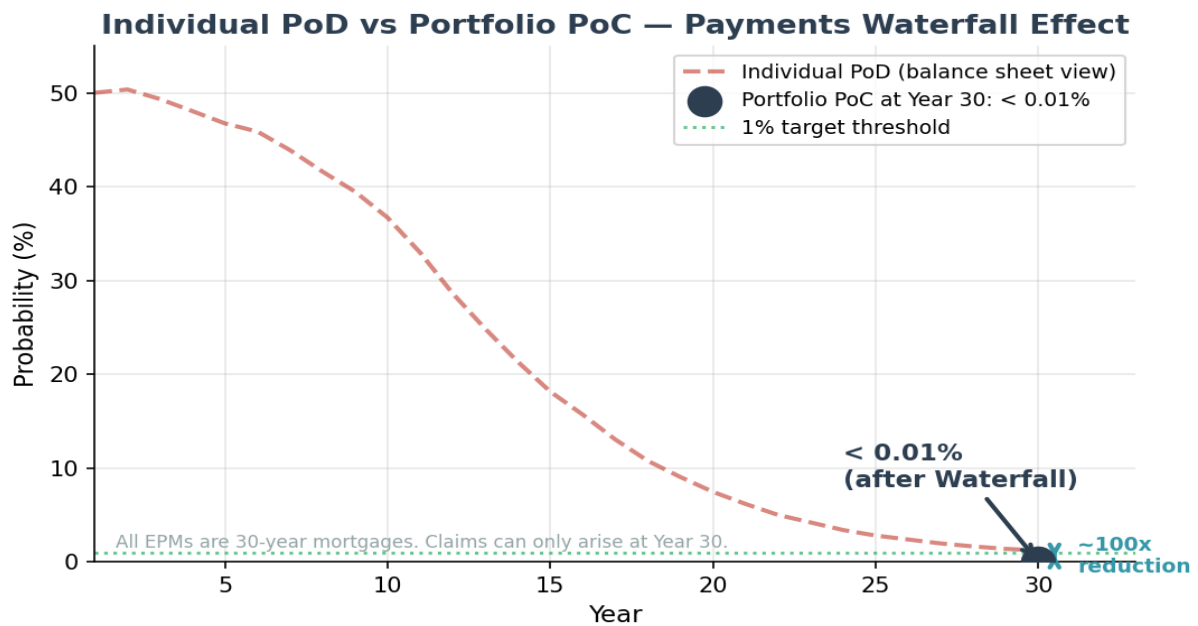
How It Works — Product Overview (v14c)

Feature	Detail
Loan Type	Principal & Interest (default)
Typical Loan	\$1.36M (62.0% LVR on \$2M property)
Investment	S&P500; index-linked ETFs via BlackRock (passive) + SpiderRock hedging
Expected Return	10% p.a. (BlackRock 30yr forward profile); net ~9.6% after hedge cost
Annuity Income	\$36,000/yr for 10 years
Profit Realisation	Every 5 years: 10% of surplus drawn
Surplus at Maturity	Split 50/50 between FutureProof and Mortgage Funder
Holiday Mechanism	Interest deferred during drawdowns (threshold 0.9/1.458)
Run-off Mechanism	No early exit until all loan costs paid from investment account
Tenure	Up to 30 years (P&I; fully amortises)
Insurance	LMI (80% of deficit paths) + tail risk reinsurance; premiums paid upfront at origination
Cash Rate	Stochastic OU (initial 4.21%, mean-reverting)
Equity-Rate Correlation	0.30 (appropriate for 30-year horizon)

The Money Flow

- 1. Homeowner takes EPM loan (\$1.36M on \$2M property, 62.0% LVR)
- 2. Calculated portion of loan proceeds held in mortgage offset account, invested by BlackRock in S&P500 index-linked ETFs
- 3. Each quarter: investment returns pay interest + costs (total ~6.65% p.a.)
- 4. Years 1-10: homeowner receives annuity income (\$25K/yr)
- 5. Every 5 years: 10% of surplus drawn
- 6. P&I; loan amortises to zero by year 30
- 7. At maturity: surplus split 50/50 between FutureProof and Mortgage Funder
- 8. No early exit permitted until all loan costs paid from investment account (run-off mechanism)

Quantitative Validation — The Numbers

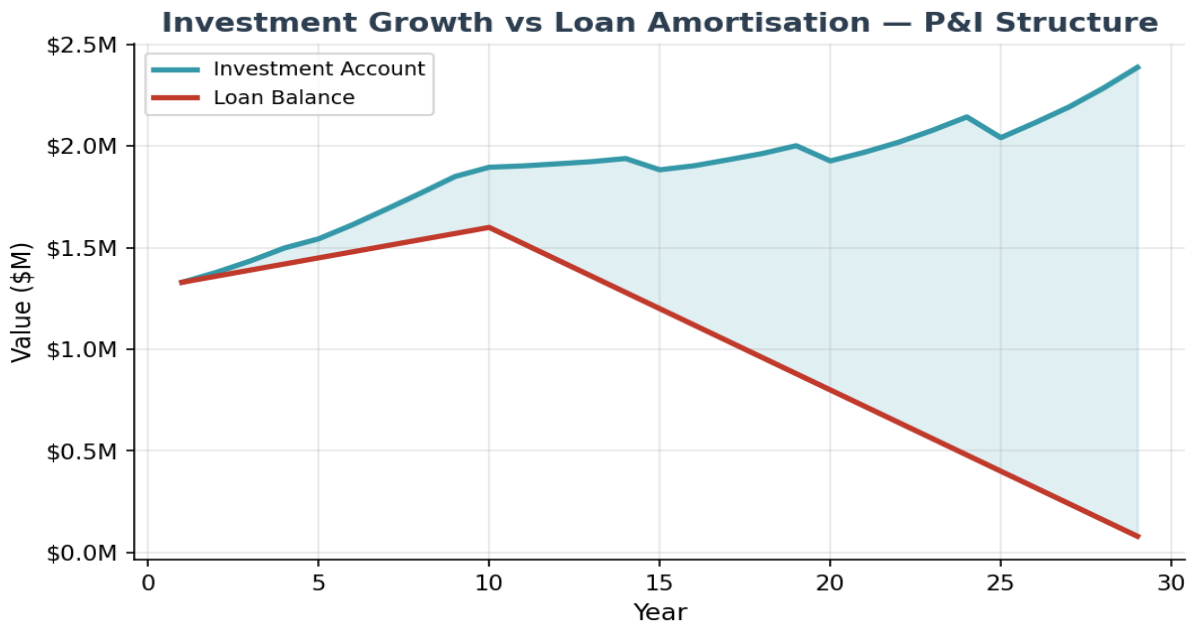


v14c Model Results (50,000 paths, P&I; + Index-linked ETF)

Metric	Value	Interpretation
Portfolio PoC (Yr 30)	< 0.01%	Probability of insurance claim after Payments Waterfall
Individual PoD (Yr 30)	1.2%	Balance sheet view before waterfall (not claim probability)
Mean Surplus	\$1,690,289	Average surplus at maturity (split 50/50 FP/Funder)
Median Surplus	\$1,376,802	Typical outcome — substantial value creation
10th Percentile	-\$211,973	Moderate deficit at 10th percentile (covered by insurance)
90th Percentile	\$4,017,354	Strong upside potential
Fair Insurance Premium (PV)	\$1,716	Based on discounted expected deficit (S3)
Insurance + Loading	\$32,929	Paid upfront at mortgage origination

Note: The key metric is Portfolio PoC (< 0.01% at year 30), not individual PoD (1.2%). PoD is a balance sheet snapshot; PoC reflects actual insurance claims after the Payments Waterfall (ETF sell-down, cross-subsidisation from surplus loans). Claims can only be made on expiring mortgages — PoD before expiry is irrelevant to insurance.

Investment Growth vs Loan Amortisation



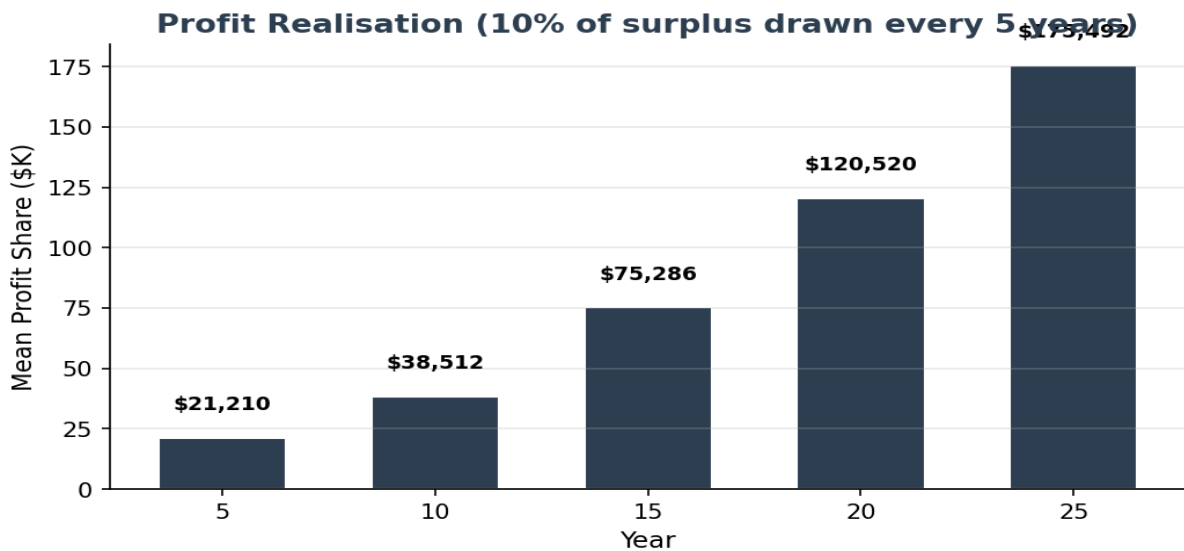
The chart illustrates the structural advantage of P&I: the mortgage balance declines to zero while the investment portfolio compounds upward. By year 15, the investment account is approximately \$2.3M while the mortgage balance has halved to ~\$675K — creating a substantial structural surplus. This divergence accelerates in later years as amortisation reduces the interest burden.

Surplus allocation: At maturity, the surplus is split 50/50 between FutureProof and the Mortgage Funder. Partial profit realisations occur every 5 years (10% of surplus drawn). House prices are deterministic in this model — they set the LVR at origination but do not affect ongoing mechanics. This is not a shared appreciation mortgage.

Revenue Model

Revenue Streams per EPM (\$1.36M loan)

Revenue Source	Estimated Annual	Over 20yr Life
FutureProof margin (0.25%)	\$3,375/yr (declining)	~\$50,000
Origination fee (1.5%, upfront)	\$20,250	\$20,250
Insurance premium (in variable costs)	Included in 3.45%	Collected with interest payments
Surplus share (50% at maturity)	At exit	~\$1,217,996 (mean, per EPM)
Profit realisation (25% every 5yr)	Variable	~\$255,528 (mean, 20yr)
Hedging program fee (0.25%)	\$3,375/yr (declining)	~\$50,000
Total per EPM		\$200,000-\$300,000+



Portfolio Economics (Multi-Region)

Scale	EPMs	Markets	Annual Revenue	Cumulative AUM
Year 1	100	AU	\$500K	\$136M
Year 3	500	AU, NZ	\$2.5M	\$675M
Year 5	2,000	AU, NZ, UK	\$10M	\$2.7B
Year 10	10,000	AU, NZ, UK, US	\$50M	\$13.5B

Competitive Advantages & Moat

1. Product Innovation

- Only equity preservation mortgage product globally — no competitor in any target market
- Integrated index-linked ETF mechanics with calculated hedging costs
- Customers retain home ownership AND participate in investment upside
- P&I; structure creates a self-liquidating product — fundamentally different from reverse mortgages

2. Quantitative Edge

- Proprietary Monte Carlo risk engine with stochastic rates, index-linked ETF mechanics, and holiday simulation
- Ability to price insurance premiums with explicit standard error bounds
- Portfolio-level modelling across 10,000 EPM cohorts with vintage diversification
- v14c model is 4th generation — demonstrating rapid iteration capability

3. Multi-Region Scalability

- Production-ready technology platform already supports US, AU, UK, and NZ regions
- Region-specific configuration deployed: currency, LTV limits, regulatory bodies, tax treatment, licensing
- First-mover advantage in every target market — no equity preservation product exists globally
- Growing demographic tailwind in all four markets (ageing populations, rising property values, retirement income gap)
- Combined addressable market of 6.8M+ households and \$15T+ in residential equity

Risk Factors

Parameter & Model Risks

- **Equity drift (μ) below the 9.2% base-case assumption.** Long-horizon valuation-based estimates (CAPE-implied 5–6%) sit materially below 9.2%. At $\mu = 8.2\%$ the model produces ~9% PoD alone; in combination with realistic γ and collar this reaches 16.9%.
- **Mean-reversion (γ) weaker than 0.163 assumption.** Bootstrap 95% CI is [0.019, 0.447] with a likelihood-ratio test p-value of 0.092 against $\gamma = 0$. γ is the single most load-bearing parameter in the model; a weaker draw materially raises PoD.
- **Collar cost understated.** 0.046% base-case input is inconsistent with Black-Scholes pricing at current AU rates (0.16–0.44%). Re-anchoring to 0.30% adds ~1pp to PoD per year of cost drag.
- Higher-than-expected wholesale funding costs — currently 2.0% in model; market comparables 2.0–3.0%.
- Regulatory changes to reverse mortgage / equity release rules across target markets.

Execution Risks

- Scaling from concept to portfolio requires significant capital
- Wholesale funding negotiations (current quotes at 3%, model assumes 2%)
- Reinsurance market appetite — sized to the adverse-plausible leg (43.2% PoD), not the base case
- Operational complexity of managing index-linked ETF positions and Payments Waterfall

Mitigants

- P&I; structure provides structural risk reduction across all parameter scenarios (loan fully amortises)
- Payments Waterfall reduces portfolio PoC materially versus individual PoD (1.2% at base case)
- Run-off mechanism eliminates early exit risk — unique to EPM, no other mortgage works this way
- Low 21% loss severity bounds expected loss even under the adverse-plausible scenario
- Insurance premiums paid upfront at mortgage origination; claims only at mortgage expiry
- Multi-region platform provides diversification across housing markets and rate environments
- Capital, reinsurance attachment, and pricing anchored on adverse-plausible scenario — not base case

EPM vs Current Investment Landscape

A critical question for investors: how does the EPM risk profile compare to existing asset classes? The following table benchmarks the EPM at both base-case and realistic-central parameters against traditional mortgages and rated credit.

Metric	EPM base	EPM realistic	Prime AU Mortgage	Moody's AA Bond
Cumulative PoD (30yr)	1.2%	16.9%	3–5%	~1.5%
Loss severity (LGD)	~21% of peak	~21% of peak	20–40%	40–60%
Expected loss (30yr)	0.13%	~2.3%	0.6–1.25%	0.6–0.9%
Ratings band (PoD)	AA/AAA	BBB/BB	A/BBB	AA

- **Base-case comparison: safer than a prime mortgage.** At base-case parameters the EPM's 1.2% cumulative PoD sits below the 3–5% prime AU mortgage range and the low severity places the product in the AA/AAA band on expected loss. This framing is correct so long as the parameter inputs hold.
- **Realistic-central comparison: comparable to prime mortgage risk.** At empirically-centred parameters ($\mu = 8.2\%$, $\gamma = 0.11$, collar = 0.30%) the same model produces 16.9% PoD. On a pure-PoD basis this is above the prime mortgage benchmark; on expected loss it is ~2.3%, well above prime mortgage EL. The product remains insurable and economically attractive, but the ratings-equivalence claim moves from AA/AAA to investment-grade-speculative (BBB/BB band).
- **Uncorrelated to traditional credit across all scenarios.** EPM risk is driven by long-term equity market performance relative to interest rates — fundamentally different from borrower credit risk. This diversification benefit is structural and holds regardless of which parameter set eventuates.
- **Funder protection is layered.** The wholesale funder holds a secured claim against real property (80% LVR), with a \$2,435,992 mean surplus at maturity at base case plus LMI and reinsurance. Under the adverse-plausible scenario, mean surplus falls to ~\$253,000 but remains positive on average; capital and reinsurance sizing is calibrated to that leg, not base case.

Summary & Next Steps

Investment Thesis

FutureProof has developed a genuinely innovative financial product. The v14c model incorporates stochastic rates, the Shevchenko mean-reverting equity process, a Payments Waterfall, explicit insurance pricing, and portfolio-level analysis. At base-case parameters the portfolio PoC is < 0.01% at year 30. At realistic-central parameters, the model produces 16.9% PoD with ~\$995K mean surplus; at adverse-plausible parameters, 43.2% PoD with ~\$253K mean surplus. The product is commercially attractive across this full range, remains uncorrelated with traditional credit risk, and benefits from low loss severity (~21%) in every scenario.

The investment case is strongest when investors understand the parameter range and are comfortable with capital sized to the adverse-plausible leg. This is the honest, defensible framing — and is how we will present the product to reinsurers and rating agencies.

Use of Funds

- Technology platform development — multi-region webapp, risk engine, portfolio management (US, AU, UK, NZ)
- Regulatory and legal framework — AFSL (AU), FAP (NZ), FCA (UK), NMLS (US)
- Wholesale funding partnerships in each market (target: 1.5-2.0% margin)
- Initial EPM originations — pilot portfolio in Australia (50-100 EPMs)
- Team expansion — sales, operations, compliance, actuarial across AU then international

Key Milestones

- 1. v14c model validation COMPLETE — portfolio PoC confirmed at < 0.01% at year 30
- 2. Finalise wholesale funding arrangement in AU (target: 1.5-2.0% margin)
- 3. Launch AU pilot program (50-100 EPMs)
- 4. Establish reinsurance partnership — < 0.01% PoC supports attractive terms
- 5. Expand to NZ (Year 2) leveraging AU infrastructure
- 6. Enter UK market (Year 3-4) — FCA licensing and local funding partnerships
- 7. Enter US market (Year 4-5) — NMLS licensing, target jumbo equity segment
- 8. Portfolio AUM target: \$675M by Year 3 (AU/NZ), \$13.5B by Year 10 (all markets)